

Iain McGilchrist ~ Active Inference Insights 023 ~ Hemispheric Lateralisation

Insights | Active Inference Insights 023 ~ Hemispheric Lateralisation, Relationality, Being | 1:59:21

hello everyone and welcome back to a very special episode of active inference insights I your host Darius parvey Wayne and today I'm absolutely thrilled to be speaking to Dr Ian mcgilchrist Dr mcgilchrist is a psychiatrist neuroscientist philosopher and literary scholar best known for his two books or rather Toms the matter in his ersary and the matter with the master in his ersary rather and the matter with things in these he explores radically different worldviews presented by the two hemispheres of the brain and the many implications both at the personal and the societal level of such cerebral lateralization in mcil Christ welcome to the show and again thank you so very much for joining me thanks Darius it's a great pleasure so I think um as I said sort of off stream plenty of people in the audience will be familiar with your work and and I'm sure have read at least one of the two books but it might be worth just starting with a very general overview of the kind of lateralization hypothesis um which we can use as a sort of launch pad to get into deeper issues yes well um most people who don't know an awful lot about it um know that this has all been rubbished long ago and that differences between brain hemis pop psychology and and all that and and that that's a position that I understand

because after some phenomenal experiments that were made possible by the so-called split brain operation carried out in Caltech in the 60s and 70s to enable people with intractable epilepsy to lead a Life by dividing the two hemispheres it was possible for psychologists to interview and engage with one hemisphere of a person at a time and get quite different results this led to a lot of speculation most of which was wrong but when we start speculating in science we usually get it wrong it's not unusual this is the whole process of science you you refine it you change it you replace it so um there are three things that make it a curious issue and one worth addressing first of all I suppose before I even say that um any neurologist knows that it's not just the site but the side of the Legion that makes the difference so the site is important but the side is crucial but when you think about it in the abstract the brain an organ which has power from making connections and whose power exists in connections is divided why it's also asymmetrical in just about everything that can be measured both functionally and physiologically why and and thirdly there is a band of fibers at the base of the brain called the Corpus callosum which connects the two hemispheres only about 2% of neurons actually cross it and a lot of its traffic is in the service of inhibition it's one hemisphere inhibiting the other as much

as exciting it so those questions led me
to uh 30 years really
of examining this question and uh what I
think uh the the explanation of this
strange phenomenon of the divided brain
is can be put like this it answers a
simple evolutionary question how to eat
without being
eaten so in order to get stuff
principally to eat but also perhaps in a
bird of TG to build a nest in an ape to
pick up something to use as a tool you
have to be able to bring a a narrow beam
sharply focused attention to Barrel on a
detail but if that's the only kind of
attention you're paying then you're not
aware of the other 357 degrees Arc of
attention available to you and in that
arena there may be a predator there may
also your kin your mate and your
offspring that you need to look after
so the one Hemisphere and in humans this
is and it becomes so over Evolution the
left hemisphere that deals with this
very narrowly targeted attention of
something it already knows in order to
grab it and get it is conducted out of
the left hemisphere it's the left
hemisphere that controls the right hand
with which most of us do the grabbing
and getting and the right hemisphere has
being left to the task of understanding
the whole of the rest of the picture so
I mean as a sound bite I sometimes say
the left
hemisphere um
manipulates what's there it apprehends
it in the sense of AD prehend grasping

onto something but the right hemisphere understands it it comprehends it in other words holds it together so those are two important distinctions and I'll just say a little about the consequences of this when one hemisphere is working alone and of course in the normal situation that doesn't happen but what we know from uh either accidents of nature or from experimentation is that the left hemisphere sees a world uh corresponding to the kind of attention it pays because attention brings into being the reality we experience how we attend matters and if you pay this peace meal ATT mention to small bits what you see is a world made up of small bits a fragmented world a world in which things are cut from Context separate from one another uh known because they're the things the left hemisphere targets what it's familiar with what it wants um they are also uh static they're fixed by this Gorgon gaze the thing I want um and they're abstract they're C ized they're decontextualized and effectively inanimate and I'm not using that expression in a loose way we know that there are differences between the hemispheres in the way in which they see things such that the right hemisphere may see things that we normally think of as inanimate as animate for example the sun and the left normally think of as animate as inanimate for example um your partner uh so uh this has big consequences the

right hemisphere meantime is seeing a world in which not that it's chaotic but nothing is ever finally fixed and certain nothing nothing is ever fully known nothing is ever entirely static it's always on the move it's always massively interconnected with lots of other things hence the context Stripped Away by the left hemisphere makes a huge difference to the meaning of what it is we're seeing um it understands what's implicit the left hemisphere more concent on what is explicit and I would argue that much of what gives value to life is implicit and this is an effectly effect a world which is um living in animous and there are just two further things to say one is that the right hemisphere sees a world as it presences uh to use a hyeran term but what it means is as the as whatever it is that we're attending to is coming into being for us whereas the left hemisphere sees a representation which is literally something that is present again after the fact when of course it is no longer present so one is effectively the left hemisphere's World rather like a map a schema a theory a diagram whereas the right hemisphere's world is more like the world that is mapped in all its richness and complexity there that was my elevator pitch in whatever it was five minutes excellent no that's that's very good uh that's excellent elevator pitch um there's a you know at least within

the active inference Community there's a
there's a general understanding or a
general uh claim at least that
neuroanatomy has a rather interesting
mapping to the structure of the world so
for example the fact that the brain is
anatomically hierarchical seems to have
some uh mapping with the the temporal T
you know the time scales at which the
world unfolds such that if you have a
predictive hierarchy let's say your
higher order um Dynamics are going to be
tracking the the slower fluctuations in
the environment and this is all this all
un plays
neuroanatomically is there a way in
which Beyond pure asymmetry and as you
say the brain is neuroanatomically
asymmetrical not only in its division uh
into the two uh hemispheres but also the
actual structure of the hemispheres and
you outline this in in the master in his
hemispher is there any other the way in
which the actual neuroanatomical
structure of the brain resembles these
underlying
functions by function do you mean the
world I I mean in so far as if we have
these dichotomized roles for the
hemispheres yes um so implicit versus
explicit the Galt versus uh the part if
you will yeah is it is there anything
that you know a neuron atomist could
look in look inside the hemispheres and
say okay there's something uh gen genely
physical that is mapping onto those
dichotomies uh indeed yes I mean I think
the structure of the brain does reflect

the if you want to put it this way the
reality that with which it's engaged
it's hardly surprising because it will
have evolved to understand whatever it
is that

it's there to engage with and of course
the question

arises is the structure of the

a function of the structure of the world
that it's engaging with or

is the way we see the world a function
of the structure of the brain and I made
that point in the presence of ran

Williams a former Archbishop of
Canterbury and he he replied with a very

Migel Christian point that I should have
forestalled does it have to be either

and I think that's probably right they
are interdependent and they reflect one
another and the thing that really is I

think not only just intellectually
interesting but of great philosophical
significance is something that I spend

the first 400 pages of the matter with
things explaining which is that when it

comes to the what I call the portals
through which we can gain information
about the world and and I take these to

be and other people might take them to
be something else but I think it covers
the ground I take it them to be

attention in the first case

perception which of course is not the
same as attention but partly depends on
attention um judgments that we form on
the basis of what we attend to and what
we

perceive um emotional and social

intelligence and understanding uh
cognitive uh intelligence IQ basically
and a degree of creativity whereby we
adapt what we're seeing according to
possibilities in each of these cases the
right hemisphere superior to the left
and and so and the odd thing is you know
well people will say well that can't be
right because why would the right
hemisphere be always superior to the
left what's the point in having it and
the answer is the left hemisphere is
vastly superior to the right in one
thing which I call apprehension which is
being able to grasp things and calculate
with them and move them about and use
them effectively so it has one value
which is utility and the right
hemisphere has other values um such as
you know goodness Beauty and truth so
there is a
hierarchy both in how good they are at
understanding and in what they're able
to understand and that
hierarchy suggests that the right
hemisphere should be if you like
in uh overall control not close control
but overall command of what the right
hemisphere is telling it and it's
interesting that from an
electrophysiological point of view the
left hemisphere communicates with the
right hemisphere less frequently and
less um uh uh to a lesser extent than
the right does with the left there are a
number of asymmetries built into the
structure and function of the brain and
they they point to differences that I

think are essential in the world
especially for people like philosophers
who are trying to um
grapple with questions like what can we
believe to be the case excellent yes so
so the notion of asymmetry here is really
interesting especially kind of
functional asymmetry which which is I
guess encoded in the very title the
master in his Emissary this kind of N
and Fable that that the right is the
master and and the left is the Emissary
um it maps at least in my mind to a kind
of classic you know you mentioned
utility there a classic distinction
within psychology and now theoretical
neurobiology about exploitation versus
exploration and these tradeoffs as I'm
sure you know that are fundamental to
our sense making and our
self-organization and um I've been
thinking about this especially in terms
of attention because attention seems
simultaneously to be a kind of
utilitarian process such that we we zero
in on um things that we want but also
it's exploratory and this comes back to
kind of the very notion of salience this
this distinction between um you know
it's Information Gain fundamentally and
I've been thinking fun is there an
asymmetry there in value and I kind of
think there is because I think there's
an interesting point that information
gain is only useful if it's in the game
of providing useful information gain to
the organism that then will uh enact
exploitative Behavior so um I've been

writing this paper it's just been accepted doesn't really matter but I have this example of um a uh a bird versus a kind of Floating Leaf in the periphery of my visual uh field and actually it isn't really worth me paying deep attention to it because to uncover the hidden cause of that is a waste of time and it doesn't yield the same pragmatic effects as it might be the epistemic value however if that object seems to start to engross my attentional field I will you know automatically turn to it and try and uncover its hidden C which implies to me that that epistemic information gain is somewhat enslaved by the utility of getting out of the way of whatever's coming my way all of this to say that is there a potential here is there a potential counterargument that the left hemisphere might itself be the master because everything is you well everything can be summed in some utilitarian sense or or in the sense of utility in terms of pure self-organization and and in terms of predictive processing and active inference providing model evidence for our preferences or just um bringing the work bringing the world that we desire into being and actually all the you know Beauty truth goodness morality and so on are in fact enslaved by that overall goal well naturally you would expect that I don't agree with that at all um it's simply the devil advoca playing you know ramachandran's Devil's advoc yes yes no that's fine of course

and we don't live to eat but Eat to Live
so the business of getting stuff to help
us Thrive is an important part of our
life but it's by no means all of our
life or the life of any animal
so I mean there is a question for the
utilitarian which was first pointed out
by Bentham in the 18th century he has to
answer the question utility For
What and if the answer is just for
further use it hasn't really got us any
further forward so utility is a service
in in permitting something else to be
going on so I mean I would say put it
from a philosophical point of view the
other way round um but I'd also say that
the one that explores which is the right
Hemisphere and is therefore able to look
around the environment and see what is
there you could say that's in utility
because I mean you can reduce everything
to a utility if you wish it's just a
point of view um but it's not the only
point of view um in that point of view
it is the right hemisphere that is being
just as useful as the left hemisphere
which is exploiting what it knows and
indeed the left hemisphere won't get
very far without having the exploratory
attention of the right hemisphere on its
own it will it will not survive and and
it is interesting that these two types
of attention and I'm sure you you were
putting it Forward um in that in
realization of that that the the right
hemisphere is the hemisphere of
exploration and the left of going a
little bit beyond what it is we know

whereas the right sorry the left hemisphere is enslaved to use your term to what it is it does know and therefore is not good at changing set being flexible suddenly realizing I need to look at this differently there's there's a big literature on that on on the relative inflexibility of the left hemisphere and when you put that together with its knowing less understanding less um I think you've got a hard case to make that and I know you're not W to making it that the left hemisphere should be viewed as the as the master there one little clinical observation is that when you have frontal damage and the posterior cortex is released from frontal control in the case of the left hemisphere the hand starts grabbing and grasping things even imaginary things but it starts to it will go to a picture and try and take something out of it it tries to lock onto if it's passing a chest of drawers it'll try to catch onto the knob whereas in the case of the right hemisphere when it's dis inhibited from frontal control you the hand starts to explore it explores the environment with sort of movement of a tentative kind going out so it is it is I mean there's no question that those are important differences between the two hemispheres quite I mean so I've been really interested in not only exploratory versus exploitative Behavior

with respect to attention but also so-called endogenous versus exogenous attention uh which is a dichotomy or distinction as you know that goes all the way back to the 80s and and posna and people like this and and and the you know to cut a long story short the brunt of my paper is I think this dichotomy is massively oversimplified and needs dismantling somewhat although the distinctions are useful but in terms of going back to the neuro Anatomy there's a really interesting finding so this is mainly Coretta and Shulman and their big 2002 nature review paper and they find that um endogenous attention which is generally taken to be go driven and voluntary is localized to the intraparietal cortex and the superior frontal cortex not necessarily with hemispheric specialization but exogenous attention uh which is localized to the temporal parietal cortex and the inferior frontal cortex is mainly lateralized to the right Hemisphere and I think this maps exactly what we were saying which is that exogenous attention is constantly being drawn by interesting and relevant stimuli one thing I wanted to ask you about is one of the classic criteria of um exogenous attention is it's involuntary and I think we we we sense that right if we hear an ambulance siren we're immediately drawn to it and this in principle is an instance of exogenous attention but that doesn't really seem to be in keeping with the right

hemisphere's
relationality um it's I don't know it
involuntary doesn't really seem to be
particularly right
hemispheric um capacity so so how do you
sort of marry that and make that
compatible well of course the first
thing to say is that involuntary
motivations are present in throughout
the brain and both hemispheres quite
obviously but I don't think the right
hemisphere's attention is simply
passively drawn if that's what
involuntary means towards a stimulus I
mean of course if there is a threatening
or exciting stimulus then the brain in
general is going to be drawn towards it
um but the the thing about right
hemisphere tension is that it can also
calibrate more easily than the left
hemisphere whether this attention is is
as it were um worthwhile is worth the
effort of attending to or not so it can
inhibit uh one of the things that is
important about the right hemisphere is
it's capability of inhibiting its own um
processes which is a greater capacity
than obviously the left hemisphere can
do that too but in in general the left
hemisphere tends to be more um again to
use your useful um term enslave
to a response to something which is its
automatic response to it and it seems um
to jump to conclusions so sometimes
people say to me so I guess that um uh
oh God what's his name you know Nobel
prize winning system one system two I've
just forgotten his name

um

sorry con I mean I I know conman well
but sorry his name just disappeared um
so caraman system one and system two and
and we've corresponded about this very
briefly but um we don't think that I
mean first of all he's carving the brain
from top to bottom more than from side
to side which I am but but also because
the evidence is that the left hemisphere
jumps to conclusions it's quick and
dirty it wants the decision fast partly
because it is the predator in us and
can't really afford to miss an
opportunity I mean it may get it it
doesn't really matter but but hesitating
and you will lose it so it tends to rush
in whereas as R chandran says the right
hemisphere is the devil's advocate it's
losing milliseconds by saying yeah but
hang on it could be this it could be
that and is much more better at
inhibiting what it is it's going to
latch on to as well as inhibiting
inappropriate emotional responses which
is uh interesting because it's well it
goes with it being emotionally more
sophistic ated in the left hemisphere
yes I wanted just the very that's all
super useful and actually speaking of
usefulness back to usefulness um there's
a constant tension I guess within the
brain sciences and the human Sciences
about the distinction between thology
and tonomy and thology is a kind of
Aristotelian notion that there really is
a a final cause and we're moving towards
something in a purposive Manner and in

tonomy is the apparent purposiveness of our Behavior behavior and I thought you said something really interesting there which goes back to lesing which is that it's always utility for something and the way that I pose that initial question was the utility is defined by these kinds of preferences which in a predictive processing hierarchy is encoded right up and down a generative model and in principle everything unfolds in the through the lens of of those predictions but I sense the kind of skepticism on your part of that because I mean one part that seems an incredibly uh selfcontained and self-referential system and doesn't really allow in any profound sense for relationality and maybe perhaps the generative model or these preferences should span the agent environment Arena but you know more broadly is there do you do you believe in a in a fundamental Telos for which are attention is being guided or these hemispheres are acting in correspondence Beyond let's say mere survival or mere self-organization well it's a very good and rather rich and complex question I think um I think it depends very much what you mean by um a telos um in essence as you pointed out um the necessity to survive must be a goal for whatever a creature is doing that that's bound to be part of it um but then you ask about whether there these

other values perhaps Beyond and I think that there are I think I think that we we've we've bought into a very hydraulic system in which what we do is push from behind so rather like a mechanism we we're sort of responding to something that pushes us and we we either push back or we go on or whatever but we're not drawn from in front and I think that when you think about the way human lives evolve and why we do any of the things we do we cannot ignore the fact that we are drawn to certain ends certain not that we have a necessary plan of how we're going to do it but those are ends that we we we feel um an Allure to that we we it's esteem we value we wish to um respond adequately to and I think there the right hemisphere is has a broader canvas than than the left in what it's doing and I'm not sure that really the distinction which I'm familiar with between teleology and theonomics is is really getting us very far I I I feel it's a kind of a verbal slight of hand whereby one can reintroduce teleology which has been a big no no for such a long time without appearing to do so and you know but at last we're able to say not that Turtles go ashore and lay their eggs but we can now say Turtles go ashore to lay their eggs they do and it seems it seems it seems particularly perverse and artificial to deny that any creature has any kind of a goal even if you look at an Amber under a microscope in in in a drop of pond

water you can see it exploring certain things and gravit towards certain things so I think that there is Direction and that direction is partly governed by our values and not the only value is utility I claim um but there isn't a kind of what people fear I think completely rightly is the idea of it's all being worked out in advance and this is a purpose that the universe is taking us to that that implies some sort of overall um very large divine kind mechanism a sort of you know Celestial engineer who's worked out the plan and I don't believe that so I think that nonetheless there is shape and meaning to the direction that Evolution has taken I think it's extraordinary that it has managed to invent the eye 14 times according to Robin Conway Morris um because it's a rather amazing thing to have invented even once yes so there's all this wonderful work by Michael Levan I'm not sure whether you're familiar with Mich work about sort of morphogen and the capacity for for these organisms sort of grow frogs legs in or an eye on the Frog's leg and and all this incred like this apparently invisible morphogenetic space in which things seem to tend um and and again I think well there's there's an ongoing debate I mean this is this podcast is called active inference in sites and often times I will diverge from active inference because my fundamental orientation is philosophy and active

inference is fundamentally aerior
physics but I'm not sure and we can you
know I can try and unpick this but
there's an ongoing debate in active
inference and and the free energy
principle which is kl's great um you
know architectural achievement which is
whether you know things purely look as
if they are maximizing model evolence as
if they're doing some basian work
they're minimizing variational free
energy so that's it looks like that's a
descriptive instrumental tool for you
and I as scientists or whether there is
you know things actually have the
architectural systems that um might
solely be descriptive so this the mark
of blanket for example or they are
actually doing some form of variational
basian inference and I guess this comes
very much to our question of Telos
andology I mean Ian is it is it is there
a risk that when I say well actually all
I can ever really say is that this is an
exceptionally good modeling tool and I
can use it to sort of get very close to
the phenomena but never really speak of
the phenomena how how much of a
limitation do you actually see of that
and and and also do you see that
applying also to the hemispheric
hypothesis yes I think one has to
distinguish
between um the of Truth and and whether
a certain thing is true to what it is
modeling uh and um whether or not it is
simply another way of describing it so I
would say that hemisphere theory is not

just another way of describing it adds something but I think it's generally true that when we model in science as Neil Bore said we are not necessarily talking about the reality of what is there what we're talking about is making our observations cohere with with um with what we making them into a coherent picture of what it is we're observing so I in other words it's not really about finally uncovering the reality it's about a way of modeling which may be good or bad and and I think that that would be the way I generally look at

it um

I I do do you find do you find that um if if the the fundamental motive is to have an internally coherent system one runs the risk of pure self- enclosure and self-referentiality um absent any ontological grounding I mean if my theory is simply that things look as if they're doing Bayesian variational Bayesian inference and I can say everything from the cosmos to an atom it is looking like it's doing so well I've got some fundamental axiom that there are things in the universe like an atom and the cosmos but beyond that it seems like I'm just constructing my theory and then building it out from the inside without ever really getting out of the cave um where it is where where is the kind of tether in your in your mind to reality I mean to to put my cards on the table I think it may be phenomena

I I think it may be Consciousness
because that seems to me to be the kind
of ontological Prior everything itself
is realized and understood through
Consciousness um and there's obviously a
massive problem here about how we can
get from mathematics to Consciousness
and going from the math to the territory
or the map to the territory but I'm I'm
curious about how you see that mapping
and and again is it Consciousness is it
mathematics what is the fundamental
tether I don't think it's
mathematics I mean mathematics is um is
of course um enormously useful and
beautiful but it it's by Nature um not
in intended to cover the whole of
reality it um I mean this was a mistake
that Galileo made when he said that the
the cosmos the book of the cosmos is
written in numbers well yes it is some
of it but also some of it's written in
things that frankly are non-numerical
and you won't understand them better by
doing more mass about them I mean there
are um people with um particularly
people with schizophrenia but sometimes
people with right hemisphere damage who
want to work out in mathematical terms
what it is that um somebody's doing when
they are falling in love or making love
or whatever and they have the ambition
to be able to reduce it all to an
abstract system of mathematics I would
say that's misguided
because our first understanding is not
that it's mathematics mathematics is
something we can bring to it but like

you I completely um would endorse this idea that it's phenomenology we we take in the world in a host of different ways and some of the ways we aim to understand it are through and this is the second part of my three-part book the matter with things epistemology how do we we get closer to truth I think one undeniably is through science other is through the use of reason but uh on their own they can't deal with everything and in fact intuition is a very important part of reasoning well I always say that people who have good intuitions reason better than those who don't and people who reason well have better intuitions than people who don't so they're they're like a pair of twins that should always work together and in a way they're in the service of intuition and this is not in any way to sort of um turn one's back on the the procedural nature of much of science but when you come to look at the way in which great steps forward have been taken in our understanding in science and in mathematics they've often come from an insight which is a gestalt shift and and from an imaginative people which is not to say that there isn't important stuff to be done at the micro level doing it in a left hemisphere linear procedural fashion yes there there's plenty there I mean one is it's kind of ironic that physicists and mathematicians think that well either the phenomena is maths or the Consciousness could be you know the the

manifestations of Consciousness could be
mathematically explained given that
their predecessors didn't think so
whether that's b or B or Einstein um
that's one rather ironic thing I this I
mean a lot of this really makes me think
of I following the master and his ersary
which I finished only about a couple of
weeks ago and and massively enjoyed so
thank you for all the insights first and
foremost I've been um I've been digging
into draus because I've always had an
interest in draus uh and and have read
around draus and a bit of draus but
never really committed to draus and
draus has this really interesting idea
that he in his um negation or his
argument against John S that activ
and it comes back to what you were
saying that we always talk about being
pushed from behind but not pulled
forward that activity has a world to
mind causal pattern such that I feel
myself drawn through what Merlo Ponte
you know would call motor intentionality
into situations and I I want to briefly
touch on draus and Mar Ponty and haiger
because I know you're interested and
extremely well read and well versed with
these people that haiger as you know
says that the the fundamental way of
being is not this kind of cartisian
Detachment rather it's doign it's being
in the
world what what does that tell us
about Phenom phenomenology what does
that tell us about our Consciousness
that if indeed it's fundamental way

of being is relational and in the world
and non rational in the sense of hyper
rationality and and unpicking things in
the left hemisphere way does that tell
us anything profound about
our fundamental way of being I think it
does yes and and you know I I I would
add to the list of philosophers you
mentioned Michael

Pani um and and also wienstein wienstein
made this um very important observation
that when you stop moving and stop
acting and just stare the world seems to
become very strange and and no longer
has any kind of meaning but that when
you act in it it does again and this is
um an

important observation and goes with the
ideas of people like Meo ponti that our
being is embedded in the world and this
comes back to my emphasis on the
importance

of uh both embodiment and context after
all the context in which we largely Act
is that of our body and our body's
engagement with the world and it's
through that that we to understand
things not by an Olympian Detachment
pretending to be a brain in a vat and
just speculating about things in the
abstract so to to understand the world
one must be engaged in it and I think
that recognition is
important one of the problems I'm sorry
I'm a psychiatrist I come back I find
the whole business of schizophrenia
arely fascinating if it weren't so as
do I

as if it weren't such a widespread condition people would be just completely amazed every time they met somebody with schizophrenia because they see the world in a totally different way and um it's not an exact match but the the simplest way of putting it is they have a hypofunctional right hemisphere in which the left hemisphere is in overdrive and so there's a kind of hyper rationalism going on which is actually not going to lead you to truth or sanity as GK chesteron wonderfully said um ad man is not somebody who's lost his reason he's lost everything but his reason and therefore has to reason on the basis of something that he doesn't understand so the schizophrenic sitting there and going somebody's talking to me in this room and there's nobody in this room so where's it coming from looks around and sees that plug socket over there must be coming out of the plug socket and that's wired into my neighbor's house and so so the whole thing is built entirely rationally but on grounds that are not based in experience so um we have to be careful about

hyper abstract and Hyper rationalistic formulations not that they can't cast light but that we should be aware that we're not actually substituting or shouldn't be substituting those for the whole of experienced reality you know the thing about the color blue and and Patricia Churchland sort of saying but the color blue just is a wavelength of

whatever it is 104 mutus per second so
but but actually it
isn't that's that's a of almost
irrelevant description of it from
outside but it blue is nothing if it is
not experiential and that of course is
not something that can be done through
um the the abstract and numerical
approach quite I mean I I feel sometimes
quite disheartened reading these neuros
um well they call themselves neurop
philosophers but it's like they never
read any David Charmers or or Thomas
Nagel but you know they know far more
than I do there's um there's so much
here and and I I'm also very very
interested in
self-consciousness um and self-awareness
and what that means because it's an
extremely misleading term yes but yes it
brings me back it brings me back to
haiger because as you know and as you've
written about haiger is constantly
tussling with this subject object
distinction so in one of his lectures he
he he says that you know the whole of
his work is to uproot the cartisian
model that has pushed you know held back
Philosophy for for 400 years
nevertheless it seems to me at least
that darine does still have this
structure it does still have this in
some ways a relationality it's not a
dissolving of things into a w like the
Buddhist might speak about there is
still the there was still at some level
the acting and the acted and yet it's
not in this sort of cartisian manner

when you think about about um self and
other um and we could I'm very very keen
to come to schizophrenia because I think
this is really a paradigmatic case where
it all becomes very interesting but just
at the sort of very neurotypical level
of being in the world what what what do
you what do you senses going on there I
mean I presume it's not some cartesian
ego homunculus floating you know behind
the eyes but what how much can we
really disband that dichotomy and how
how important is it to still keep
something like a subject object
distinction

yes well the first thing to say can I
start from the end of that question and
go back to the first part of it um
however you yeah
um if I can remember uh
so I think the the ability to hold
distinction without vision is critical
to understanding and it's something we
seem to have lost so if you make a
distinction people run very quickly to
the being a complete
divide whereas it isn't it's rather like
you know the two poles of a magnet they
are very distinct neither can live
without the other but they are not
divided in fact they are completely
joined and you know so and if you try to
cut one off from the other you just
create another the magnet which is like
a hologram of the first so the the idea
this idea turns out I think to be really
it recurs throughout all my
philosophical Explorations the need for

Union and division to be unified not to be divided so Gerta said the whole business of nature is dividing the United uniting The Divided and the Cy and the deast and so on he is a marvelous piece but effectively he kept coming back to this that we should see something that it seems one to be manifold but see the manifold as being unified at some level and and you know in in the matter of things I have a whole chapter on this idea which is so important in eastern thinking of the relationship between the one and the many so I'm so I don't want to say that before saying anything about division and and Union in the case of self and other and I I think in the I think it's in the master in his Emissary I I give the example you know of the infant that is completely fused with the mother shortly after birth and there is a total inter subjectivity and that the business of the the gradual separation from the mother over the first year is negotiating a distinction which is not of severance so it's actually an it's not a diminishment of something it's an increase in it in much the same way I sometimes use the image of um two two heavenly bodies that are rotating around circulating orbiting around one another and and if for them to have that orbiting relationship they mustn't crash into one another but they mustn't just fly apart and indeed you can say that about a human relationship that it can

become over and meshed so that the the
self of each party is not preserved or
it can become
so divided that the two have to part so
the the whole thing is about holding
this tension between two things without
collapsing it into one or the other into
mind or body into self or other but
holding the tension and that negotiation
which as I say is what happens in the
properly functional relationship through
the Intercourse between the right ventor
medial frontal cortex of the mother and
the right ventor medial frontal cortex
of the of the infant brings about this
ability to be sort of together but not
um fused and I think the distinction
between the objective and the subjective
is a very important one but the idea
that the objective as naively supposed
that it cuts off every kind of
human contribution to the
understanding has got to be wrong
because we we can't understand things by
not being a human as it were that we'll
understand them less well from a very
peculiar point of view in which
everything about us has been cut off
that won't lead to a real or true
version of what it is but what I think
that objectivity means is the ability to
inhabit other perspectives on your own
so objectivity is really seeing things
from as many perspectives as you can and
that's very valuable but it doesn't
negate the fact that you have your own
point of view your subjective
position yeah well I think um the work

of more modern phenomenologist like like
a Dan Sahari for example or a Sean
Gallagher would suggest that the
very Act of empathy in this very broad
sense of being able to take the
perspective of another is
Downstream on the fundamental for meanus
um what people call pre-reflective
self-consciousness such that you know my
experience is for me and I have a
privilege access to my own experience
and in doing so I recognize that you do
as well and there is this Mutual
modeling and shaping um and yet and and
yet um and and so inter subjectivity is
a component of this kind of subjectivity
and and yet it there still seems to me
at the fundament
a division is the wrong word and I don't
want to say division but the fundamental
fact that Consciousness experience is
always for me and not for you um and
this goes back to all the way back to
sort of Louis S and P and the work they
did with Dan sahav on the fact that in
schizophrenia even that famus is still
not lost so for insertion is a really
useful example here um you know metser
talks about the fact that there's
there's no sense of agency over that
thought or there's no and others have
talked about there being no possession
over that for but that for is still for
the subject whatever that whatever the
form that subject takes so given that is
there um is there a way in which
by the very n by a very nature as
conscious beings who have a privileged

access only you know only to our own
lens to our own perspective we are cast
off from others in in some in in the
deepest manner
possible I don't think that no
um so Consciousness is
Consciousness is for me best imaged by
by water um
so this is an image from shelling but I
think it works for me that in the water
there can be a Whirlpool or many
whirlpools and and the first thing to
notice is the whirlpools are not in the
water they are the water they're not
sort of an extraneous object in the
water they're just the form the water
takes at the time and those World Pools
May be very self involved so that the
movement all which returns within this
this
flow um it's very real it can be
photographed it can be measured it can
have power to move stones and it will
eventually move on and then it becomes
part of the rest of the stream now I see
Consciousness the individual
Consciousness as being somewhat like
this that it is connected and in
an enormous number of experiences it's
hard to dismiss all of them as
delusional people seem to be able to
feel their Consciousness as part of
something greater or part of can
communicate with somebody else so I I
think that I wouldn't take a hard and
fast view on Consciousness as it's
always just me because I would want to
you know problematize the word me what

exactly am I I am part of I am part of
first of all um the society uh that
formed me the the culture that formed me
the culture to which I hope I contribute
I'm also part of nature nature is in me
it's not around me like the environment
it's in me and I'm in it and I would say
because we don't have to go there at all
but but I believe that actually very
importantly there is um a Divine or
sacred aspect to the cosmos and that in
that we also exist so I I would you know
take take a different view as if were
from the kind of atomistic view of the
eye yes I mean there's um again it was
it was merely a a proposal I don't quite
know where I land on it because I'm I'm
also quite taken by the work in this
case of of SRA and um more laterally
Kenneth willerford on the fact that what
we call subjectivity may just be
conscious and just knowing s so um
Kenneth wilford's got this really nice
paper I think 2016 in one of met
journals about how you know
Consciousness is this bizarre phenomenon
where there is this phenomenal feeling
that it's for a subject but every every
time we've tried to find that subject
whether it's neurophysiologically or in
the humean introspection we can't find
it and so what does it mean to be for a
subject that itself might not be present
and he says that well that the subject
relatum of Consciousness might just be
the Consciousness episode itself yes and
I think that's um a potentially quite
neat solution yes to this kind of vexing

issue but more broadly about
Consciousness is the river and I really
like I I too enjoy this metaphor because
it also brings to mind this this
relationship between change and Oneness
and and brings to mind herac ex his
insights makes me think firstly about
it's consciousness's relation to matter
because I is it would you say well
matter is by nature somewhat
atomistic um self-
distinguishing and cut off and it seems
to me that you know the
whirlpool has uh it although it's part
of nature it's part of the water it also
has this kind of self-d distinguishing
way of being and so is there a way in
which Consciousness interacts with
matter such that in me it's taking this
form in some intricate manner that I I
can't quite understand because I don't
know the relationship between matter and
Consciousness and I'm sure no one does
but in some ways matter is fixing
Consciousness at least for a a transient
period of time such that it can manifest
in a in a thing like me hello all uh
before letting Dr Migel Christ answer my
question I wanted to very quickly jump
on and let everyone know that when I was
formulating this question what I had in
mind was actually a talk that uh Ian had
given about three years ago on the
relationship between m own Consciousness
and the reason why I want to make this
very clear is because it might seem that
in formulating that question like it's
my idea it's my theory and I want to

express that it's not and it's totally my fault for not having referenced the talk and not having made it clear that actually this was Ian's uh original Theory and idea uh the reason why I formulated this question is because obviously a lot of people might have not seen the talk and I wanted to lure and get I to talk about this incredibly interesting idea uh but I think unfortunately the execution it may have seemed that I was trying to take credit for it and I'm really really not so I wanted to come on before the podcast goes live because the last thing I want to do is take credit for ideas that aren't mine plagiarize especially when it comes to someone like Ian who has worked so hard for so many years to establish these incredibly rich and insightful and thought-provoking theories so all of this to say that this notion of uh matter as the kind of stabilizer facility Ator um yeah of Consciousness is not my theory and so if it came across like that when I was asking this question I really wanted to just come on the record and say uh it's absolutely not it's Ian's what I will do is I will link this wonderful talk that he gave it's about an hour and 50 minutes it's on YouTube I will link it in the video description below but anyway the reason why as you'll see Ian basically recapitulates what I've said is because obviously these are things that he had already spoken about uh so yeah totally my fault but wanted to come

on here and clear that up because again
I don't want anyone shouting at me I
don't want anyone getting the wrong idea
and I definitely don't want to be taking
credit for ideas that aren't mine so
with that being said I hope you enjoy
the rest of the conversation and I will
now leave you with Ian because he is far
more interesting and wise than thank you
what I think about the problem of matter
and Consciousness as we think of
it is that frankly
despite enormous efforts over very long
periods of time nobody has been able to
give a coherent and
convincing explanation of how something
like Consciousness can simply emerge
from matter unless matter somehow
already partakes of Consciousness or is
in Consciousness or is an aspect of
Consciousness if it has nothing to do
with Consciousness then no amount of
hand waving and gesturing towards
complexity can really make something
like Consciousness the inwardness of
things come out of
it um and In the comparison between how
we know Consciousness and how we know
matter Consciousness is primary in that
we can only know matter via
Consciousness I may only know
Consciousness via matter the brain but I
don't know that it's certainly not
necessary but it is necessary that to
know matter
um and by the way we've never seen
matter nobody has ever experienced
matter all we have experienced is

aspects of our Consciousness that have qualities that we call material and I think that what matter is is a phase of Consciousness much in the sense that water goes through phases it's flowing and transparent and runs across your hand it's opaque and hard and can crack your head open it's in this room there's tons of UN believably in this room and if there weren't I wouldn't live and I can't see it so conscious people say well it doesn't look like matter well no it wouldn't do but um it doesn't have to what matter is doing I think is providing two things one of them is permanence for a while a degree of resistance which brings permanence so it's resisting change and that brings permanence and it's also resisting is containing and providing some sort of bound to what I can do and what I can think and without any bounds without any um the thing becomes chaos and without any definition for creativity to happen there must be an opposing force that is overcome and anything to be capable of lasting it has to have this degree of resistance to change and that is what matter is in my view wonderful yes there's so much here um that comes to mind I mean very briefly I guess a materialist neuroscientist here would might pulling their hair out and going well just because you can't know ever there's just an epistemological limit that means we can't ever know for sure

that the brain creates Consciousness
that needs to be held as distinct from
an ontological premise that the brain
causes Consciousness so it might be I
never know this I can never prove it
because of David Chalmers and Nagle and
and all of the philosophical arguments
that have been leveraged in this
direction but nonetheless um on on
logically speaking it's the best
solution we have um that they seem so
coupled and the brain is so complicated
and it's doing all these wonderful
things of course Consciousness just
Springs out because the moment you shut
one part of the brain off or you have a
stroke or a lesion uh cons some element
of Consciousness disappears do you find
any truck in that uh distinction between
the epistemology and the ontology here
not really because um perhaps of VI will
disappear if I change the settings on my
screen um parts of experience will
disappear if you if you go around
injuring the brain no no problem there
the but I don't show I don't believe I
believe there are only three
possibilities frankly that the brain is
either an emitter of Consciousness which
seems to be the the favored one amongst
um reductionist materialists or it is a
transmitter of Consciousness or it is a
permitter of Consciousness in other
words it somehow shapes and filters
Consciousness and makes it my
Consciousness and that's what I believe
the brain is and I know you weren't
making an argument there but you gesture

towards complexity but you know as
you're perfectly aware the the
cerebellum has 80% of the neurons in the
brain and is massively more complexly
interconnected within itself with pinji
cells being one of the
most um massively ramified and
interconnected cell neurons and and yet
it doesn't sustain Consciousness on its
own so I I think it's not like that I
mean I had a conversation with um Mark
SS the other day and and Mark says you
know there's this yes place in the in
the in the tectum I think maybe the
tegmentum but I think it's the tectum of
the midbrain where you know two cubic
millimeters of white matter if you put a
needle in and and disturb it then the
person's gone but I don't think he
thinks and he said he didn't that that
little bit of matter creates the
Consciousness it can't do well he's
Mark's been on the podcast before and I
like Mark's I like Mark's work a lot and
I think it's very insightful I guess
he's a he'll call himself a dual aspect
monist um and so there's some underlying
thing and Consciousness is one aspect of
it and matter is another it seems to me
that you're leaning slightly towards a
more pansus position but it it's almost
a different form of pans sism from let's
say a traditional
um well I guess beron Russell and his
monism it comes to mind here because
Russell and laterally Philip gof who's
the kind of premere um panus at the
moment this idea that well as you say if

I take um you know it's very akin to what you're saying about never having seen matter if I have this cup of water um it's not that the matter is the cup of water right its fundamental nature is consciousness and matter might just describe what um what physics does but not what physics is now is there a difference there between what you're saying which is the conscious matter is a facilitatory thing or process rather um that allows for creativity and the resistant forces that lead to self-actualization versus saying well no no what matter actually is is consciousness is are those two different things well I'm naturally a bit skeptical of grand pronouncements like that mat is consciousness but I think it's an I think it's a way in which Consciousness can manifest um and perhaps that's not a huge difference um I'm having a conversation with Philip go go as well so I'm interested in what he will he will have to say about my view but my feeling is that he and I um probably come quite close on this so yeah it is kind of have you much about sorry on yeah no no no I mean again to lay my class well I'm quite a part I'm I'm quite party to a pany but that might just be because I I don't know think yeah no I I look I I guess I don't know oh one thing I do push back against is the epistemic hubris of thinking that we know um and we

certainly don't what have you thought
much about Hydro
enic children and what that might say
for this relationship between matter and
Consciousness because these are children
who's uh such an enormous portion of
their brain is absent and yet they seem
to
be behaving in a conscious manner now
again a materialist reductionist might
say Well they're not conscious and um
they might employ some kind of zombie
argument what what what do you think of
the kind of ramifications of that well I
I would again you know
reject epistemic hui here and say I'm
not sure but but um we to make it clear
for um any viewer the what we're talking
about is not hydris in which there is
quite a large part of the cranium filled
with cerebrospinal fluid instead of with
the brain tissue that would normally be
there but I think you're referring to
hydran and ke in which children are born
with virtually no brain at all so things
stop at the brain stem that's right um
and what is quite extraordinary and I
don't know how to explain it is that in
the absence of any kind of um visual
cortex they are able to
recognize and apparently see and
distinguish people in their faces they
like pieces of music and have preferred
pieces of music they have preferred
toys um
and I think once you've seen one of
these children I mean of course nobody I
mean you can be dayart and say I don't

know that what I see out of my window
are really people they might just be
machines walking about with hats and
coats I mean it's always open you to
take that position but I want to appeal
to a degree of intuitive imaginative
common sense about
it that it's very unlikely that that
that is the case it doesn't feel like
that and
I wonderful photos of
them these photos of them sort of at
Disneyland smiling while their while
their siblings are on their lap and um
yes I think Common Sense might pril but
again humility first and foremost
there's a um the one thing I wanted to
ask about when you're talking about
resistance is the importance of negation
in your work yes um so you I think I
believe you quote Spinoza who says all
determination is
negation and I think you say that um
there are two things one can do in this
world is either say no or not say no
which I think is fantastic um because
the word yes just doesn't even intervene
but what does that again going back to
sort of Consciousness and our way of
being how does that play out in terms of
our experience what what is it is
this a metaphysical fact that in terms
of things uh self- individuating there's
is a as you say a fundamental tension
that if if the you know you know if if
became LAX would there would we would
just dissolve into the soup and active
inference for reasons like we can talk

about agrees with this or or is this
also about our conscious experience that
in some ways that which means that it
doesn't dissolve into the Super and just
becomes about everything intends about
everything is the fact that
Consciousness in some ways is saying no
and also permitting things to enter into
its
field yeah
um I suppose what I was getting at
was
that first of all negation is how
anything becomes defined as what it is
after all defining is laying down
boundaries and in a way something only
comes into being through the being
bounds to it if it is
boundlessly mixed with everything else
there's only one thing it can be which
is God or the Eternal being or something
but anything that we know in this world
has to be defined and this process is
creative it's how you know when a when
an artist is putting down paint on the
canvas he or she is making a decision to
paint this and not something else and
it's a very simple point that you know
when Michelangelo created his D David he
didn't put anything together he just
said no to the stone all over the place
until finally he'd got the thing that he
wanted um and you could say he was
saying yes to the the image I suppose
but I suspect that from what one knows
about the Act of Creation that the thing
actually becomes what it is in the
process and it is an evolving

relationship between the artist and the
artifact or the work of art
so and what I was suggesting is that
when we see things
seeking I give the image again of water
because it's so useful for so many
things but the image of the river you
know apparently seek its way through the
landscape but all that what is happening
is that the landscape is either saying
no or not no to the water it can't say
yes to the water it's not in a position
to do that it either fails to stop the
water or it stops the water those are
its choices um but that actually brings
into being the river and I think that
this is an image of
how um almost anything comes into being
that in the in the first instance
it's through the absence of negation to
something and let's see where this goes
and what happens and one the thing is I
suppose um achieves a certain degree and
there isn't an absolute cff Point um of
actualization then it is more possible
for us to respond to it as it is and
wish to further actualize it yes again
it brings to mind draus um because draus
has this idea that sort s of when one is
uh actualizing action you never have the
the conditions of satisfaction in your
mind you're just it's just
unfolding and I well but interestingly
draus does talk about and I believe this
is based off Merlo Ponte that there's a
kind of tension in that act and I think
he uses the word tension and by that he
means um there's a if one senses that

one is moving away from an optimal grip
on the situation one is always drawn
back to to what you know to an expert
way of being um and again I think this
just it really does speak to the fact
that there is no conscious affirmative
yes in that it is just sculpting away
and and and and again it comes in some
ways back to Aristotle I mean
Aristotle's idea that you can only judge
a life once it's been lived it's it's a
disclosure it's an unfolding it's not um
it's not purposive in the sense of a
kind of cartesian
removal and this is what I want to do
and I'm going to do it now and no this
is this is this is not quite right and N
said um the purpose of a Melody is not
to reach its final note but unless it
reaches its final note noted as a Melody
so yeah I mean I'd love to go back if we
can to self-consciousness because I
think it's so important and it also
leads to schizophrenia in which there's
a kind of hyperconsciousness that sty is
the person's existence in a way but what
you know I don't know why but when you
were giving those instances I was
thinking of something I describ in the
matter with things which is to do with
expertise again draus is topic that in
the early stages of learning expertise
it's useful to have things to guidelines
to follow rules to observe but that when
you get to the the last two of his five
stages of
expertise having such rules or
principles is actually damaging it

limits your ability to develop your skill and you remain forever mediocre and one of the things I'm intrigued by is people who have motor skills such as I I I I out of the blue the The Physician who looks after the um the TT races on the aisle of man you you'll know what I'm talking about but um maybe for the International audience there's this completely crazy and infinitely dangerous sporting activity that takes place once a year on the Isle of Man off the coast of Britain during which all the islands roads are closed and motorbikes race around a 62 mile course that goes over ordinary public roads um round tight bends over Braves and you know down sudden inclines and so forth and at certain times theye they they're speed can approach 200 miles an hour so this is both very dangerous given this is not a perfectly smooth track um and um remarkable but the thing that comes out of it is that they enter a sort of mental state in which they have a kind of total awareness but if anything distracts their attention to a detail then they they lose the grip and a lot of it is to do with not trying so you know and this is a familiar story isn't it that one Rider um felt he was doing very well and then he realized that something was happening to his bike and he had to stop and fix it and he stopped and fixed it and after that he just poled on because he thought there's no point in in going fast now because I know I've lost the thing actually he

turned out to have the winning speed
so well there there's a there's a
paradox in there I think this is
Slingerlands trying not to try um and
I'm not quite sure how to resolve that
Paradox yes but it's fascinating on I I
hate to I hate to do self aggrandisement
and self advertisement but actually very
fitting we're talking about flow States
because I just uh myself and several
others including KL freston just posted
this uh just published this paper on
Flow States under an active inference
framework um and self-awareness in flow
States uh which was just published a
couple of days ago in Frontiers so you
may you may find that to be um that up
but perhaps perhaps slightly frustrating
we try we try and provide a kind of
basian model for FL dates and and
perhaps draus would be screaming at me
because he'd be saying oh you can't
mathematize it um but really interesting
something that's very interesting about
dra's skill hierarchy he has this
example about airport um about fighter
jet pilots and that when fighter jet
Pilots instruct novice fighter jet
Pilots there's a checklist of uh you
know an order so you must check the
light and but when they actually I
believe it was Ice scan experiments on
fighter jet pilots in the Jets they
realized they were not following that
order whatsoever so it's not just that
the rules are um embedded in coded
unconsciously followed out it's almost
that they're not followed out which

implies that it's um they in expertise
perhaps and maybe this is not the case
for all expertise but draus is quite
strong that chess play expert chess
players are not following rules um
there's almost a relinquishing of rules
and there's there's actually quite a
strict division between those who have
to kind of follow this epistemic
Behavior this guideline following and
then those who just do in some sense and
again I don't I don't think D has comes
to the conclusion of exactly how that
works but it's very interesting to think
that this isn't just a uh absorption Act
of of absorbing rules but actually might
be an absolving of rules it might be a
den of R I'm not quite sure what that
means very it's fascinating none
interesting I mean I hope that what goes
on a lot of the time is the absorbing of
procedures so that you're not even aware
you're doing them but you're you're
suggesting and and have no reason to
doubt it that at times they're not
followed um I mean it comes back to the
observation that you have to really
really know the rules before you're
allowed to break them and I think that
is very important because um too much of
the time we neglect skill in in in um
allowing people to they want to develop
a talent and what we do is encourage
them just do what you feel like and let
it all hang out and you know I know
where that comes from and some of that
is good but actually something very
important is also much more important

which is you really do need to learn the procedures you know by the time Raphael was nine he could master all kinds of techniques that most people would be drawing for decades before they'd be able to do so but he did actually learn them he didn't just wasn't born with them so there's that

um but yes I mean

when when that pilot landed the plane on the Hudson and when a surgeon suddenly reacts to whatever is going on in theater I mean when asked afterwards they often say well I don't know what I did I just did what I have to and and it is interesting that that may actually involve denying certain rules I mean obviously there are certain rules you know why they're

there and in certain and in most contexts they're important but in a certain particular context it might be a crucially waste time wasting step to follow that rule or procedure and you need to cut through to something else

yes well it speaks to our our bureaucratic infrastructures which I'm sure we we can speak about at length I mean one thing I will will add is that we we don't at least in this paper we don't deny the the presence of anything we we don't just act as if um you know well there would be no model if it was just this kind of elusive mysterious unfolding this disclosure I think what the really important thing that we we tried to emphasize was context and the importance of contextual cues and so we

argue that a lot of what is happening in
flow States for example is this
constantly unfolding contextual queuing
which we sort of can understand as a
habitual mental action and draf talks
about this a lot as well that even if
it's not okay uh given this contextual
cue I do this it puts you in this kind
of gestal condition where you know
exactly what to do without knowing that
you know and it's all implicit and it's
unfolding and yet if exactly what you
said about the distraction if a
distraction is sufficiently uh takes you
away if it doesn't act as a contextual
cue for this unfold these unfolding
Dynamics then you're out of the state
and you have to work your way back into
it by in some ways
recovering quite a loaded philosophical
term these contextual cues so I think
all of this is very much um on on on
track and I think we're finding
convergence here I guess another part of
this is intuition yes and the role of
intuition in in in skillful coping and
expertise what do you what do you take
intuition to be and how much do you
think that it drive lives our everyday
Behavior because as drers and Mer Ponte
and Hy would tell us we live intuitively
right I know how to sit on this chair I
know how to speak English I know how to
drink water is there um how how can we
integrate intuition into this picture
yes well again I'd sort of reverse that
and say how do we integrate the rest
into the picture of intuition which is

the the overarching
picture I mean if you think about
intuition is basically the unconscious
processes that we have we can now put a
fairly accurate figure on this that all
that our nervous system and our body in
general is aware of um
is infinitely greater than the amount
that we self-consciously are aware of so
the little bit we're self-consciously
aware of that you know that you know is
the bit that you're looking at with the
left hemisphere uh focal attention but
that is um estimated to be slightly over
half a percent out of the 100% the rest
is all being done uh outside the glare
of attention and you know sorry this is
a bit kind of um simplistic I guess but
I hope you'll forgive me um I think
people say the unconscious in a lot of
people's minds it's something like the
cellar it's somewhere down there as
it were in the dark that is you know a
tank below everything that's going on in
the sunlight and occasionally fish pop
up from the bottom tank and occasionally
boat go down but instead I would use the
image of a a spotlight on a stage so
where the spotlight is is what you're
focused on but it hasn't altered the
rest of the stage all the rest of the
possibility is still there it's just not
actually currently in the spotlight and
I think that's a better way to think of
it now what intuition is really saying
is
that what it means is to
contact that other

95% in whatever way and it's not inferior I mean of course it can be mistaken but as I um repeatedly say um simply following a line of argument can lead to mistakes as well as in the schizophrenic for example so you need context and you need intuition as well as you need the power of logic and not missing any element in there so what I think intuition enables us to do is the complex stuff so it enables us to make decisions with far-reaching consequences to balance alternatives to um to to to form an intention to do a certain thing we don't do these things consciously they come up through periods when we are not attending and what they have over the explicit Consciousness is that in that explicit Consciousness if I articulate what I'm thinking we can only have a sentence or serial sentences but in intuition there may be 12 or 15 strands that are contributing and when I try to articulate I automatically go back to the obvious I I articulate what seems to be Salient but all the things that are not Salient but nonetheless very important are speaking to me with the result that I can't remember his name but there is an academic jurist at one of the max plank institutes in Germany um who says he's the director of it and he says people who are in charge of large organizations should be telling their people to use their intuitions and I think he's dead right I mean there's an army of um psychologists no disrespect to the profession who are

going around making a lot of money
telling people in organizations not to
trust their intuitions but I think this
is very very much a half story and the
worst half of the story because by not
listening to your intuitions you end up
in very strange places I think there are
examples around us of how that is
happening so in the matter with things I
have a whole section of three chapters
on intuition and the literature on
intuition and how it works and whether
it works and so on and once again um
it's not an All or Nothing thing but the
when we overthink things we usually make
them worse

unless unless things have already broken
down in other words it's like a toolkit
that you get out of the boot when you
need to repair and this is where
Whitehead said this wonderful thing that
operations of thought are like Cavalry
Charges in battle they can only be made
at decisive moments they require Fresh
Horses and they they be the main thing
you rely on so this is this is how it is
we shouldn't be thinking that by
overthinking we will make better
decisions we often

[Music]

don't well I I think to add to that um I
complete I'm in complete agreement I
think the other I slightly ironic thing
is this claim that you could ever
dispense with intuition um and and I
think if you try to dispense like this
is I think in many ways it comes to what
you were saying about the the take that

the non-conscious is a kind of uh um
discarded Sellar that very rarely sees
the light and I think in some ways we
can blame Plato for this and some Two
Worlds mythologizing but my my my take
on this is that for to use your
Spotlight example and as you know it's a
very common example in the attentional
literature it's often taken that this
this Spotlight is is s it's moving
itself it's like the the hand is within
the spotlight my claim is that no no no
this Spotlight is fun Mally enslaved
again to use this word by so much that
you can never become aware of so even
the very fact of paying attention to a
siren or it is not an arbitrary act to
pay attention to a siren even if it
means even if it's purely reflectional
why is it reflective well it's probably
because at some point there was some
evolutionary pressure to pay attention
to things that are loud or luminous and
whatnot and I think there's been a great
mistake in dichotomizing these things
exogenous endogenous unconscious
conscious such that we don't realize
that they're always in play with one
another they're always in concert and if
you think that your Spotlight is solely
uh guiding itself you are very blind to
how biased that Spotlight is and and how
much it is driven by that which looks
underneath your conscious awareness so I
think I think again it's it's it points
exactly to what you said in all of your
books which is
relationality that it's there is no one

the other and in many ways a self-guiding spotlight would be totally useless to what end apart from the spotlight own maintenance would you know it has to be in some ways in constrained and subserving and and being a very good Emissary but I worry that the spotlight attention metaphor is itself emblematic of this kind of self- referentiality that the left hemisphere is yes is propagated and you know one of the things I ought to just mention is that people worry that if um things happen through unconscious promptings that somehow they are not deciding but this is because they've already decided that they are just that little tiny bit of conscious them as though all their unconsciousness was not also a very important part the the main part of who they are and it it's you know that's the right right yes well again it's the great irony that even that part of you which is conscious is not blind or uh encapsulated with respect to the parts that are non-conscious they're constantly filtering information in and out of one another so even if you wanted to say oh I'm just this little moral that's well you can't be because it's kind of like heraclitis is River it's always in flux um and excellent okay this is all super I want to very briefly ask about this the relationship that we've been speaking about with respect to the hemispheric differences and and and the underlying cognitive

infrastructure to broader social changes
historical changes um so when I was you
know when I finished the master in his
emory I I like reading book reviews
after reading books because I I'm always
interested to see whether my thoughts
converge with with other peoples and I
thought there was an extremely um
slightly shallow one by Owen Flanagan
um so this is 15 years ago so you may
not remember he
he he even gets some
the he gets
it yes he gets it mixed up but he makes
see he makes the claim that um 2,500
years of History are ill understood as
our hemispheric differences and then as
a result trying to do the mapping
between hemispheric differences and
historical evolution is a slightly fra
problem I'm curious about just generally
and it could be brief what do you what
relationship do you establish there
between the brain and historical events
whether it's the Greeks the Romans The
Romantics and so on I can't help
reflecting that um Flanigan's University
Duke just invited me back to give the
John P McGovern
lecture I looked for him in the
audience he was hiding no um there can
be a misunderstanding because I mean
very literal-minded people go but a
brain is something in a human being it
can't be in a society and you know okay
I get that but what I'm pointing to is
not either that changes in the brain
have caused changes I I I'm not in the

business of saying what causes what and
I don't think the brain itself is
causative of such changes in society but
what I what I thought thought about it
was really inspired by Lou Sassy's work
particularly Madness and modernism and
you know this extraordinarily brilliant
book I think in which he looks at the
patterns of thinking and creativity and
art in the last 100 years or so and uh
relates it to the phenomena experienced
by people with
schizophrenia and obviously it's not
that we've all got schizophrenia it's
that as he hints in one of his
appendixes
um schizophrenia mimics what it's like
when you have right hemisphere
dysfunction and so what he's really
saying is at the moment our society
ignores all the kinds of more subtle
things that the right hemisphere knows
in favor of fatally simplified
theoretical understandings of things and
that this is causing problems so I
thought of thought well H so but
presumably this well clearly this hasn't
always been the case so when looking
back in history what do we see and and
you know in a in a former life I was
effectively interested in the history of
ideas and um so I I thought well let's
review Western Civilization from the
Greeks through the Romans up to The
Rance and and Beyond to the
Enlightenment and Romanticism and
modernism and what I think I can see
there in three cases is that there's

been a a change in emphasis um over time
uh in which in each case there is
initially a rather sudden and
surprising um period it tends to happen
rather quickly in which suddenly there's
a move forward of both the right and
left hemisphere together working well
complementing one another always with
the left hemisphere being the as it were
the functionary that assists the master
rather than taking charge that's when
things go astray but as as each
civilization has advanced in terms of
its its wealth its power its extent this
has um gone hand in hand with an
extension of the left
hemisphere's um control because the left
hemisphere's resold DET is in fact
control Power utility um and as result
the subtler things that had sustained
the civilization began to wne and the
civilization crashed so what I'm talking
about there is is not that I mean
obviously the brain is always Evol
evolving so there will be changes if you
were to brain scan a Kean man in the 6th
Century BC and and and somebody now but
they'll be very minor and they're not
what I'm talking about what I'm talking
about is how we use our brains our
brains are are there still and are right
hemisphere is telling us things but
we've been somehow
indoctrinated and I'm not suggesting
that there's some you know cult that's
indoctrinating is I'm just meaning the
mindset that we've got in through
through an industrialized Society

and a capitalist

Society is that everything is to do with
the massing power and material it's it's
all about utility and that everything
else is sort of going to hell in a
handcut effectively um but it's not that
we lack the Insight so what a lot of
people who read my work either the ma
master or the
matter um say to me um is you you made
me aware of things that I knew at some
level but
had no way of articulating or or or you
know arguing for but but you've given me
that yes I mean that I mean I think
that's the great import of your work is
again it comes back to this idea of
um of revealing that which was already
there so I I never when I so I haven't
read the matter of things uh and I will
but when I read the master and his
Emissary I didn't I mean you you you're
obviously introducing a hemispheric
theory I don't I never felt that you
were imposing anything I never felt it
was an imposition on my way of
understanding it was more of a um as you
say this kind of um this unraveling this
unul you know the way the liit talks
about knowledge is it's just chipping
away to to leave something that was
already there I mean I think an
interesting um Revolution that's going
on moment is both mindfulness and
psychedelics um and to be uh oddly
personal I'm in Mexico currently and
Mexico is uh particularly replete with
Sil cybin and um on a Sil cybin trip a

couple of weeks ago quite an intense one
and this may in part be because of the
my reading of your work it was this
remarkable multiplicity and unity unity
multiplicity but I wonder whether I
wonder what whether what you thought
about this which is to what degree will
the people who are reading your books
who then go out and and see uh you know
the universe in a in a petal to
paraphrase William Blake um to what
degree do you worry that it's this is
just a sort of biasing effect of having
read your work and it's not actually a
kind of authentic disclosure of the
world's structure at least on their part
but rather a sort of again a kind of
left hemisphere Eric self- evidencing
self-confirming uh way of being because
that was in part my worry over the
experience it was incredibly profound
and beautiful and I was in the middle of
the forest and things were felt very
much alive and and and had their own
autonomy and yet were partaking in this
kind of broader structure but after it
finished I thought well maybe that's
because I've been reading a lot of Ian's
work um and so is there a way of
potentially mitigating this this uh this
potential
well I have nothing to say as you know
about um mind altering
substances of this
kind
um it's interesting though that you
think there was
some cousinhood between whatever it was

you were reading and this experience I like that and I don't know what I think about that I mean people often say is aren't these substances sort of releasing the the right Hemisphere and I don't think they are I think what they're doing is they suppressing frontal

cortical uh function and the frontal cortex of course is the major filter of what it is that we do attend to and experience and I think that that's when that's removed then there is as the flood gates are open and and you see a lot of things together and maybe you get a sense that yes they belong together which is I think probably a real feeling that they do we can't go on leading our lives in that state whereas I hope that we can go leading our lives with um whatever it is I've given um to people because they do often say things like you've altered my life once you see this you you can't not see it and and they say for the better you know I they report moving things about their way their lives have changed through reading my work which which is completely unforeseen I had the slightest idea I hope it would be interesting to some people I had Great Hopes but had no idea that it might actually have this effect on people which is just totally lovely um obviously I I I don't think that what I'm suggesting is is is um is false or farfetched I think if it were probably it wouldn't have the impact that it generally seems to have

even on people who are fairly skeptical to to begin with you know because most people in in particularly in the world of psychology I think are there a sort of Catch 22 that when you're told that hemisphere differences are not important um because they don't really mean anything then of course you don't do the reading to find out whether they are or not so it's a self-confirming situation and one of the things I'm battling against is um the the tendency to fail to report um hemisphere differences I'm hoping that in future when people do brain research they will always even if if it's a null finding you it just report are there any hemisphere differences it's a question that really must be asked and the more we ask it the more we will understand what I've tried to understand no doubt inadequately but you know we will understand it better well you're your I mean the master his Emy and I'm sure the matter of things is obviously replete with references nonetheless so people can go there but I I I agree I mean I think in many ways as a kind of budding psychologist and philosopher it's it's as you said it it's a consequence of the Frameworks in which we adopt and so if we have a particularly popular framework and let's say something like um top down versus bottom up differences and everything is studied or or or or pursued through the lens of hierarchical systems and as you say for example caraman and tki very

interested in in in this verticality and
you know to the to the detriment of
horizontality and well I I I think hope
y you'll you you'll start to see the
fruits of your labor um because I think
I do think the conversation is around
hemispheric differences is is is
changing quite immensely talking about
Gran unifying theories I asked people uh
on Twitter very briefly a couple of days
ago about what they wanted me to ask you
okay and um you know this is as I said
it's it's an active inference Insight
podcast although I spend so much of my
time talking about Consciousness it
ought to be a Consciousness podcast but
people wanted to know about what you
thought about this kind of grand ifying
Theory and whether you'd thought about
it at all so have you given Carl's work
much reading uh have you spoken to Carl
and have you got any IDE you know any
thoughts on the free energy principle or
or predictive process processing or
anything you know aligned well I feel
that I'm slightly like molier's cook in
in this you know molier when he was
trying out his plays he used to read
them to his cook first to find out what
the reaction was um and I'm afraid I'm
as ignorant as B cook and I have I have
tried to understand and I I hope I have
understood some of what the active INF
principle involves and the free energy
principle I I'd like to be told really
what what does it add to something that
we have always believed which is that no
perception is naive that we don't we're

not just like sort of photographic
plates but we go to the world with um
predictions with presumptions about what
it is we're likely to see and so on and
that's of course very
beneficial um it not only saves energy
but it saves time and it makes us much
more effective at
um living but what what does this really
add to that because I mean yes we we
change our behavior in the light of that
p and the behavior makes the environment
feedback I mean really what it what it
seems to me to be saying in a rather
more complicated way is what I'm saying
is that everything is relational that um
what we what we perceive and take to be
reality some of it comes from us some of
it comes from out there these that
doesn't make it unreal and that the
environment if I must call it that but
the world around us and ourselves are
interdependent we are bringing one
another about so there's a constant
dialogue between them that's the way I
see it and I find that the the the mass
and all that is very impressive but I'm
not sure quite what it is um perhaps you
can tell me what it is it's adding I
want to say so Carl what is the question
to which the answer is the free energy
principle okay so if I if I can speak on
his behalf which I can't but I shall I I
think I think I actually think it'll be
a wonderful conversation with you with
you and car and others but the idea here
is that well there's an ongoing debate
is what I'll start with there's an

ongoing debate between the instrumentalist and realist depictions of active inference so on one hand you say okay we can model the behavior of everything through this basian framework so everything that persists through time as the thing that it is can be described in terms of basian inference and what this means is is that things embody in code a model of themselves and the world and their relationship and they act or they can be described as acting in order to provide evidence for that model and this can be taken from um this as I said the SC I think this is the one of the important things of this is a totally scale-free principle so you can leverage this to describe the existence of and the persisting existence of anything from an atom to the cosmos now that's the instrument of this thing which is it's an extremely useful modeling tool because it talks about how things can be sparely coupled to one another and thus how they change in relationship to one another exactly what you've been talking about such that if I act on the world and the world acts on me we enter into this mutually coupled econ Niche agent relationship which again is No not not different from the insights of ecological psychology or um embodied fory cognitive science now the other the the the the complete other end of the spectrum is the realist version of Act of inference which in many ways takes a predictive processing account which is that the brain cannot is not

just that it can be described as
encoding a generative model it is a
generative model and the way all the
brain cognition is or all forms of brain
cognition are fundamentally
predictive excuse me such that we
leverage
predictions we leverage predictions
about the world and when the world when
we receive the data from the world which
is ultimately always Divergent from our
predictions we either update our model
of the world or we act to change the
world so as to um make that sense data
more concordant with our generative
model and so in principle this is a
updating of Helm holian um percept his
notion of uh perception as inference But
it includes action and I think that's
the great part of it which is that um we
in many ways Enslaved the World by
changing it in accordance with our
preferences now obviously the world is
also enslaving us right because if a car
hits me doesn't matter what I want from
the world that car has provided evidence
for itself as a kind of bulldozing
mechanism but I think what a lot of
people are interested um from you is
again like the maths is the maths and
people are trying to understand exactly
how um this can be implemented
computationally I guess the question for
you is what is the role of a kind of
meta Theory or or or were the
limitations of a meta Theory and as you
say is this just a a reording of
something which was implicit in all

these other strands of theoretical
biology Neuroscience physics and so on
or is it actually saying something new
so I really think ultimately this is
just telling me that you should speak to
Carl um well I'm sure it would go but
I'm curious about you very Molly cook
here
um I I it I've given you Theon Pub
version fine I mean that's maybe maybe
what I need
um I mean I I still feel that I'm not
aware of what it is that it's added to
my my understanding of how perception
and action go on uh it hasn't changed
the way I think about it maybe because I
haven't understood it or maybe because
I've already intuitively understood it
and Incorporated it I don't know but um
the thing that it does do though for me
is to bring with
it a feeling that comes from the sort of
overwhelming idea of this meta Theory or
meta world or meta
whatever that
somehow somehow we've mastered this we
can predict it it's all
mechanical it it produced an image of a
world that I don't recognize and I don't
really want to know whereas I'm more
Blake like I go out and I see these
wondrous living beings and and all that
and the beauty of the world and the
complexity and the unpredictability of
it seems to have been lost in a sea of
graphs and sort of you know calculus I I
I not that there's anything wrong with
it I would want to say but I think it

needs to recognize its own partiality
that it's just a partial yes approach
that's what I'd say about the
mattera thing and I don't claim by the
way that you know
well no I just going to say no no please
I don't claim that the hemisphere theory
is a sort of meta Theory though I
believe it has many many consequences if
it's true and the evidence for those
consequences is something that I'm you
know looking for both in the um
scientific literature and in the social
psychological literature
yeah well I guess one thing to add to
this is the active inference or the free
energy principle cannot well some people
might think it can but it can't solve
Consciousness and so um we're still left
this massive but I guess I guess to to
to finish up to return to the
hemispheric differences you finished the
the the master in his Emissary with this
with this very like provocative also
quite uh alluring and endearing line
which is that um if it all turns out to
be a metaphor then that's fine because
you have high regard met which I liked a
lot and metaphors really um underpin for
me the they underpinned the the tenor
the the the frust of that whole book and
and and again holding this tension that
Mets allow us to understand the world
but also um in many ways if if it is
just a metaphorical Theory we still need
to pay respect to that and also return
back to the phenomena yes what um just
in terms of the hemispheric differences

what I I was slightly confused by what
it actually meant for that to be a
metaphor what would it mean for the
hemispheric theory to be a metaphor well
what it what it wouldn't necessarily
mean is that um there's anything wrong
with the science um Science is based on
metaphors um the important point is that
you can understand anything without a
metapher and um that that the um the use
of metaphor in science and philosophy is
is particularly um important and heavy
so you cannot step outside metaphors if
you wish to understand something it's
very obvious in in physics and it's been
argued for by various philosophers um so
I think that's what I mean I meant there
it was a slightly Sly one in which I was
kind of I was giving a little obeisance
at the end and saying well maybe I got
it wrong but nonetheless actually you
can only understand through metaphors
and I've given you a pretty important
metaphor but but rather literal-minded
people have gone oh well oh so he thinks
it's only a metaphor like only a
metaphor meaning there isn't any literal
truth to it but there is literal truth
to it I mean I'll I'll you know stake a
lot on that and I've taken I hope you
will well you know what you do with your
time is up to you and it's a terrific
ask to read this book that matter with
things but I think after this I'm
certain you will enjoy it but in it I
take the the discussion to Greater
levels of dabs and you think I've put a
lot of evidence into the master in his

chemistry and the bibliography there looks like a a a postscript um the bibliography here is 180 Pages because I really needed to look at a lot of stuff and I really did look them up and inspect them because a lot of the time it's amazing working in in in the area is that you look up something that's been quoted and requoted and quoted as saying X and you look them up and actually it says not X it doesn't even say why not so there are one or two which that has happened but uh yeah so I do care I care about empirical evidence I think it's incredibly important and if we lose that what have we got you know to I don't say empirical evidence is everything of course it has to be taken up into an understanding and so forth but if we actually throw away the importance of empirical evidence we're cast free we have no loadstone to to guide us and one of the thing Sinister things that that's happening now you wanted to talk about what's happening now very briefly but I mean one of the things that worries me very badly is the well-known fact that science is now being heavily skewed by political pressures to conform to a certain narrative about the way things are and that um perfectly good science that suggest that it's more complicated than that is not being recognized and when Nature actually writes an editorial saying it's not going to publish things just because they're good science they

have to be on message well that's the
end of science as far as I'm
concerned do you with that in mind do
you hold you know people talk about the
pendulum the the political pendulum
swinging and I guess in this case it's
the the scientific pendulum swinging do
you think nature and I mean nature both
you know both in its literal sense of
Nature and and nature is in the journal
will will nature bite back at nature you
know will will will this will will this
come home to R because I think there is
this uh utopian I guess there's this
utopian idea that if we can be if we can
self-censor if we
can confirm the theories that we already
have in mind without questioning the
underlying axioms of those theories then
everything will just be Dandy do you
think um that reality itself always
comes home to roost and we bite back
well they do if we give it long enough
to do so without destroying ourselves I
mean in the longer run reality will come
back of course and utopianism has been
the cause of the most cruel despotic
totalitarian projects in human history
so no we must resist that at all costs
and and I suppose what is beginning to
happen is that people are starting up
universities private universities where
people can really learn and study
without being forced into a straight
jacket about what they say and journals
will spring up that will publish things
and so on and no doubt Twitter will ban
them or something but we've got to get

Beyond this kind
of mind that we we ban things if we
don't like them the answer is not that
the answer is yes to be able to look at
what is said in a reasonable rational
temperate Manner and discuss it because
if we could do that there is nothing
that cannot be
resolved discussion is better than
fighting I profoundly
believe well indeed I think um again
this is I think a nan or Doan idea that
you can't get rid of things by pushing
them underground in many ways you just
make them more
vicious um yes and curious inverse way I
saw that with religion in the Soviet
Union um I've some I mean although I I
suppose I'd have to call myself of a
religious cast of mind I've always had a
kind of LoveHate relationship with the
church but when I was in Russia in the
Soviet years in the 70s I remember going
to the uh cathedrals in what was then
Leningrad and Petersburg and seeing how
they thrived you know middle of
Wednesday morning on a weekday the the
cathedral was packed with people praying
lighting candles singing together and so
on I thought this is this is so bloody
moving because also as they came out
there were these leather jacketed
sunglass thugs photographing them all so
that they could make sure that their
children would never go to university so
it took courage you know and when you
repress it it thrives now that is no
longer repressed I don't know I've heard

that congregations are falling
off maybe wrong well that there a return
to a I don't like the word spiritual but
a um a spiritually minded way of being
right I mean in many ways one has to
give credit here to to Someone Like
Jordan Peterson um and I know you've
spoken to to John vvi I think John and
you and John Share quite similar
perspectives here with a skepticism on
organized religion but also recognizing
the loss that a fundamental
reality and being 25 years old and
having grown up in Metropolitan London
my whole life um I can tell you that the
urge in me is to be extremely cynical
and condescending towards those of a
religious mentality because it's always
been the way I was raised you know but I
think I think there's a there's a
counter and whether this and I guess the
interesting point is the metaphysical n
grounding of this is some what well it's
not arbitrary but it's it's it's it's
not as important as the fact that church
or um these centers are points of
communion I know and I I think we
dispense with that our our Peril I mean
I found for example that actually team
sport is the most
unbelievable um form
of healing in some ways just to just to
be in concert with people to be trusting
people to in flow with people and um yes
I I think I think this actually all
points to your work that uh as you say
relations precede relata that we are we
are kind of nothing

without

relationships this is well this has just been an absolute Joy I am so very grateful to you I'm very grateful for the Mexican WiFi for having only messed up once um I'm on the border of Mexico I'm on the border of Mexico in Biz and I have a boats to take to bise of all places so the Wi-Fi there I think will be particularly spotty so I'm um I'm very grateful for time um I always like to ask people to just say where people can find them uh find their work I mean obviously you have these two enormous and epic and wonderful toes but beyond that where can people find what you're working on current I mean what are you working on currently well uh you won't find that anywhere actually because I'm not really sure

um I've got a number of number of things that are bubbling away um I may not have time to finish any of them but um basically the place to go is channel mcris if you just um put channel mcris into the Google bar it'll take you there um some people think it's an imperative but it's just a description and uh there's uh YouTube you can always just Google my name on YouTube and I'm afraid there's an appalling number of interviews on there my friends complain that they can't go anywhere i' well I'd like to I think no you're one of the great podcast guests thank you I don't I I I genuinely do mean this um some of the podcast no well some of them you've done are just

wonderful um I particularly liked your first one with Kurt J mongle yes where you're sort of guiding him his slightly uh yeah he he it was incredibly intense now he's he's read the matter with things we're planning to do another so that's good but he wanted to make sure he he'd read it properly well good well he's he's also got some overlap with the active inference Community actually because he's interviewed Carl a number of times and um I think he he hosted some round tables so so the worlds do Collide and converge but um I'm going to read the matter of things um I'm going to make that I'm going to make that a project okay very good um let's see how long it takes no hurry okay thank thanks very much thank you hly thank you enjoy being in Mexico it's a lot warmer than it did here on Sky I imagine but good luck yeah I've heard it's rather chilly on the British AIS so I'm I'm quite happy despite their electrical issues all right byebye wonderful all right so much bye thank you